



Project Title: TOUCHTASH – Intelligent Salon Booking System

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1. INTRODUCTION

The beauty and wellness industry has grown significantly, but many salons still use manual booking methods such as phone calls, appointment books, and walk-ins. These methods often cause scheduling conflicts, long waiting times, and poor customer communication.

TOUCHTASH is an intelligent full-stack salon booking web application designed to modernise appointment management. It allows customers to view services, check stylist availability, and book appointments online through a simple and responsive interface.

The system is developed using HTML5, CSS3, JavaScript, Bootstrap 5, Node.js, Express.js, and MySQL (Workbench). It follows a client-server architecture and uses RESTful APIs, with secure authentication implemented using JWT and bcrypt.

This project applies key Internet Programming 2 concepts such as frontend and backend development, database integration, API design, and responsive UI development.

The main goal of TOUCHTASH is to improve customer convenience while helping salon owners manage bookings and operations more efficiently through digital technology.

2. PROBLEM DEFINITION AND CONTEXTUAL JUSTIFICATION

Many local salons and beauty establishments continue to depend on manual appointment booking methods. Customers are required to phone the salon, visit the premises physically, or simply walk in and wait for an available stylist. While this process has been used for many years, it presents several operational and customer service challenges.

One of the major issues experienced by customers is excessive waiting time. Since appointments are managed manually, clients often arrive without knowing whether their preferred stylist is available or how long they will need to wait. This results in frustration and wasted time.

Salon owners also face difficulties when managing appointments manually. Tracking bookings, customer preferences, stylist schedules, and cancellations becomes increasingly difficult as the business grows. Manual scheduling frequently leads to double bookings, missed appointments, and inefficient utilisation of staff.

Another challenge is the limited online presence of many small salons. Modern customers increasingly expect businesses to provide online services, including appointment booking

and service information. Salons without digital booking systems risk losing customers to competitors that offer greater convenience.

During peak business periods, manual booking systems become even less effective. Staff members spend valuable time answering telephone calls and recording appointments instead of serving customers. This reduces productivity and negatively impacts customer satisfaction.

Stakeholders Affected

The primary users affected by these challenges include:

- Students who require affordable and convenient salon services.
- Working professionals with limited time who need to schedule appointments in advance.
- Community members seeking reliable beauty and wellness services.
- Salon owners who require efficient business management tools.
- Stylists who need organised appointment schedules to maximise productivity.

Proposed Solution

TOUCHTASH addresses these challenges by providing a comprehensive online salon booking platform that digitises the appointment management process.

Customers are able to browse available services, check stylist availability, schedule appointments online, receive personalised recommendations, and manage their bookings from any device.

Salon administrators gain access to a secure management dashboard where they can monitor appointments, update stylist schedules, manage services, and analyse booking information. This reduces administrative workload while improving customer satisfaction and business efficiency.

3. PROJECT OBJECTIVES

Main Objective

The primary objective of this project is to design, develop, and deploy an intelligent full-stack salon booking web application that simplifies appointment management for both customers and salon owners.

Specific Objectives

The project aims to achieve the following objectives:

- Design and develop a responsive and user-friendly frontend interface using HTML5, CSS3, Bootstrap 5, and JavaScript.
- Implement server-side functionality using Node.js and Express.js.
- Develop RESTful APIs to facilitate communication between the frontend and backend.
- Integrate MongoDB Atlas as the primary database management system.
- Implement complete CRUD (Create, Read, Update, Delete) functionality for users, bookings, services, and stylists.
- Develop a rule-based recommendation engine capable of suggesting suitable salon services based on customer preferences.
- Secure the application using JWT authentication and bcrypt password hashing.
- Ensure the application is responsive across desktop, tablet, and mobile devices.
- Prepare the system for cloud deployment using modern hosting platforms.

4. PROPOSED SOLUTION

TOUCHTASH provides an integrated digital platform that improves salon appointment management through automation and intelligent features.

Customers can create user accounts, browse available salon services, select preferred stylists, and book appointments at their convenience. The booking process includes real-time availability validation to minimise scheduling conflicts and improve service reliability.

The application includes a comprehensive service catalogue featuring hair styling, colouring, treatments, nail services, skincare, facials, and makeup services. Customers may filter available services according to category and personal preferences.

An intelligent recommendation engine analyses user preferences, such as hair type, face shape, and occasion, to recommend appropriate salon services. This provides customers with personalised suggestions similar to those offered during an in-person consultation.

Salon administrators have access to a secure dashboard where they can:

- Manage customer accounts.
- View and update appointments.
- Manage stylist availability.

- Add, modify, or remove salon services.
- Monitor booking statistics and business performance.

The system is fully responsive and optimised for desktop computers, tablets, and smartphones, ensuring accessibility from any modern device.

Overall, TOUCHTASH improves operational efficiency while providing customers with a convenient and modern booking experience.

5. TARGET USERS

TOUCHTASH has been developed to serve multiple categories of users within the salon industry.

Primary Users

Students

Students often require affordable and convenient salon services that can be booked around their academic schedules. The online booking platform enables them to secure appointments without visiting the salon beforehand.

Working Professionals

Working professionals generally have limited free time and require flexible appointment scheduling. The system allows them to reserve appointments in advance and receive reminders of upcoming bookings.

Community Members

Members of the local community benefit from easy access to salon services, online booking, personalised recommendations, and improved customer service.

Secondary Users

Salon Owners

Salon owners use the administrative dashboard to manage appointments, monitor business performance, supervise staff schedules, and maintain customer records.

Stylists and Beauticians

Stylists can view daily appointment schedules, update their availability, manage booking requests, and access customer service histories to deliver more personalised services.

System Administrators

System administrators oversee platform maintenance, user management, database integrity, and overall system security to ensure reliable operation.

6. SYSTEM FEATURES

TOUCHTASH provides a comprehensive set of features designed to simplify salon operations while improving the overall customer experience. The system offers different functionalities depending on the user's role within the application.

Customer Features

Customers have access to an easy-to-use interface that enables them to perform various booking-related activities. These include:

- Secure user registration and login using JWT authentication.
- Browse available salon services by category.
- View stylist profiles, specialities, ratings, and availability.
- Book appointments using an online booking system.
- Reschedule or cancel existing appointments.
- Receive personalised service recommendations based on customer preferences.
- View previous bookings and appointment history.
- Update personal profile information.
- Access a personalised customer dashboard displaying upcoming appointments and recommendations.

Administrator Features

The administrator dashboard provides complete control over the management of the salon.

Administrative functions include:

- Create, update, view, and delete bookings.
- Manage customer accounts.
- Add new salon services.
- Edit service descriptions, pricing, and duration.
- Remove outdated services.
- Manage stylist profiles and working schedules.
- Monitor appointment statistics and business performance.
- Generate reports showing popular services and customer activity.

Stylist Features

Stylists have access to their own dashboard where they can efficiently manage daily appointments.

Available features include:

- View daily, weekly, and monthly appointment schedules.
- Update availability status.
- Accept or decline appointment requests.
- View customer booking information.
- Access previous customer service history.
- Update completed appointment status.

These role-based features ensure that each user interacts only with the functions relevant to their responsibilities.

Component	Techology used	purpose
frontend	HTML5	Structures the webpages
frontend	CSS3	Provides styling and layout
frontend	Bootstrap 5	Creates responsive and mobile-friendly interfaces
frontend	javascript	Implements interactive functionality
backend	Node.js	Manages routing, middleware, and REST APIs
Backend framework	MYSQL	Handles server-side processing
database	MySQL Workbench	Stores user accounts, bookings, services, and stylist information
authentication	JWT	Secures users sessions
Password security	bcrypt	Encrypts users password
API communication	RESTful APIs	Enables communication between the frontend and backend
Version control	Git hub	Supports collaboration and version managemtn
deployment	Render	Host applications online

Technology Justification

HTML5 and CSS3 were selected because they provide a solid foundation for modern web development.

Bootstrap 5 was used to create responsive layouts that adapt automatically to different screen sizes.

JavaScript enables dynamic interaction between users and the application without requiring page refreshes.

Node.js and Express.js provide a fast and efficient backend capable of processing multiple requests simultaneously using asynchronous programming.

MySQL Workbench was selected to manage the relational database because it provides a structured approach to storing customer information, bookings, services, and stylist data while supporting efficient querying.

JWT authentication ensures secure user access by generating encrypted access tokens after successful login.

bcrypt protects user passwords through one-way hashing before they are stored in the database.

GitHub facilitates collaboration among group members while maintaining version history throughout development.

8. SYSTEM ARCHITECTURE

TOUCHTASH follows a three-tier client-server architecture that separates the presentation layer, business logic layer, and data layer. This separation improves maintainability, scalability, and system performance.

Presentation Layer (Frontend)

The frontend consists of responsive web pages developed using HTML5, CSS3, Bootstrap 5, and JavaScript.

This layer is responsible for:

- Displaying user interfaces.
- Collecting user input.

- Validating form data.
- Sending requests to the backend.
- Displaying server responses.

Business Logic Layer (Backend)

The backend is developed using Node.js and Express.js.

Its responsibilities include:

- Processing client requests.
- Authenticating users.
- Managing business rules.
- Handling appointment scheduling.
- Performing CRUD operations.
- Generating intelligent service recommendations.
- Communicating with the database.

Data Layer (Database)

The application stores its data in a MySQL database.

The database maintains information relating to:

- Users
- Stylists
- Services
- Appointments
- Reviews

The backend performs SQL queries to retrieve, insert, update, and delete records whenever users interact with the application.

Intelligent Component

The recommendation engine is integrated within the backend.

After customers enter their preferences, such as hair type, face shape, and occasion, the server evaluates these preferences using predefined rules and recommends suitable salon services.

This intelligent component improves customer satisfaction by providing personalised recommendations during the booking process.

The TOUCHTASH database has been designed using a relational database model to ensure data consistency and efficient management.

[illegible]

Navigator

SCHEMAS

Filter objects

- mutict
- sakila
- sys
- touchashdb
 - Tables
 - barbers
 - bookings
 - hairstyles
 - reviews
 - services
 - users
 - Views
 - Stored Procedures
 - Functions
- world

Administration Schemas

Information

SQL File 5 touchashdb.barbers barbers bookings barbers users barbers bookings hairstyles x

1 • SELECT * FROM touchashdb.hairstyles;

Result Grid

id	style_name	face_shape	hair_type	description
1	Taper Fade	oval	curly	Modern fade for curly hair
2	Buzz Cut	square	straight	Simple low-maintenance style
3	Pompadour	round	wavy	Classic stylish haircut
4	Mid Fade	oval	wavy	Clean modern fade
5	NULL	NULL	NULL	NULL

hairstyles 1 x

SQL File 5 touchashdb.services services

1 • SELECT * FROM touchashdb.services;

Result Grid

id	service_name	price	duration_minutes	description
1	Haircut	120.00	45	Standard haircut
2	Beard Trim	20.00	20	Professional beard trim
3	Fade Cut	100.00	60	Modern fade haircut
4	Kids Cut	90.00	30	Haircut for children
5	Hot Towel Shave	180.00	45	Classic straight-razor shave
6	Hair Colour	350.00	90	Full hair colouring
7	Shampoo & Style	70.00	25	Wash, blow-dry & basic styling
8	Beard Sculpt	110.00	35	Detailed shaping + oil treatment
9	Kids Fade	100.00	35	Fade cut specially for children

services 1 x

SQL File 5 touchashdb.users users

1 • SELECT * FROM touchashdb.users;

Result Grid

id	full_name	email	password	role	created_at
1	Test User	test@test.com	\$2b\$10\$4A2s6UmvUxhcPCuSQPZV.../ppQxZrPub...	customer	2026-05-31 0...
2	Asanda Nhleko	technicalprogramming285@gmail.com	\$2b\$10\$qLzlo8nEJwOovOcdMRWesPe39EWWuY...	customer	2026-05-31 1...
3	Luyanda Zulu	sdevelopment98@gmail.com	\$2b\$10\$gsWV1NAIdWki9bq4f6CjqqeFN54/AJx6...	customer	2026-05-31 2...
4	Asanda zulu	a_admin@supremebarber.com	\$2b\$10\$47qqgvMcw.WQ3aP4sGgqUeJLarC177...	customer	2026-06-10 1...
5	Nqubeko Gumede	brokenasanda@gmail.com	admin123	admin	2026-06-10 1...
6	Admin User	admin@touchash.co.za	123456	admin	2026-06-25 1...
9	New Admin	newadmin@touchash.co.za	admin123	admin	2026-06-25 1...
10	Asanda Zunqu	asandanhleko1@gmail.com	\$2b\$10\$/JnWuta.AOwqsaRfltu6OOe4dmLXNE...	customer	2026-06-25 1...

10. REST API DESIGN

TOUCHTASH uses RESTful APIs to facilitate communication between the frontend and backend. Each endpoint performs a specific function and follows standard HTTP methods.

HTTP method	endpoint
POST	/api/auth/register
POST	/api/auth/login
GET	/api/services
GET	/api/stylists
GET	/api/bookings
POST	/api/bookings
PUT	/api/bookings/
DELETE	/api/bookings/
GET	/api/recommendations

API Communication Process

1. A user submits a request from the frontend.
2. The request is sent to the Express.js server.
3. Authentication middleware verifies the user's JWT token when required.
4. The appropriate controller processes the request.
5. SQL queries interact with the MySQL database.
6. The server returns a JSON response to the frontend.
7. The frontend updates the interface dynamically without refreshing the page.

Using RESTful APIs ensures that the frontend and backend remain independent, making the application easier to maintain, extend, and deploy in future versions

11. IMPLEMENTATION REQUIREMENTS (FULL-STACK INTEGRATION)

The TOUCHTASH system follows a full-stack architecture that integrates the frontend, backend, and database into one complete web application. The project adopts the Model-

View-Controller (MVC) design pattern, which separates application logic into different layers to improve maintainability, scalability, and code organisation.

Backend Implementation

The server-side application is developed using Node.js and Express.js. The backend is responsible for processing client requests, performing business logic, communicating with the database, and returning responses to the frontend.

The project structure is organised as follows:

Models

Models represent the database tables and define the structure of the application's data. They include tables for:

- Users
- Services
- Stylists
- Appointments
- Reviews

Each model interacts with the MySQL database through SQL queries and ensures that data is stored consistently.

Controllers

Controllers contain the business logic of the application. Each controller performs specific operations such as:

- User registration and authentication
- Booking creation
- Appointment updates
- Booking cancellation
- Service management
- Recommendation generation

Separating business logic into controllers keeps the application organised and easier to maintain.

Routes

Routes define the available API endpoints that the frontend can access.

Examples include:

- Authentication routes

- Booking routes
- Service routes
- Stylist routes
- Recommendation routes

Each route receives incoming requests and passes them to the appropriate controller.

Middleware

Middleware performs operations before requests reach the controllers.

The application uses middleware for:

- JWT authentication
- User authorisation
- Input validation
- Error handling
- Logging requests

This improves both security and application performance.

Asynchronous Programming

Node.js uses asynchronous programming to process multiple requests efficiently.

The TOUCHTASH application makes extensive use of **async/await** to execute database operations without blocking other users.

For example, when a customer books an appointment, the server:

1. Validates the booking information.
2. Checks stylist availability.
3. Inserts the booking into the database.
4. Returns a success response to the frontend.

This approach improves system performance and provides a better user experience.

Frontend Integration

The frontend communicates with the backend using the JavaScript Fetch API.

When a customer submits a booking form:

- The browser collects the user's input.
- The information is converted into JSON format.
- A POST request is sent to the backend API.
- The backend validates the request.
- The booking is stored in the MySQL database.

- A JSON response is returned.
- The frontend displays a confirmation message without refreshing the page.

This seamless communication creates a smooth and responsive user experience.

12. INTELLIGENT AND ADVANCED FEATURES

One of the primary objectives of the TOUCHTASH system is to demonstrate intelligent web technologies. To achieve this, the application incorporates data-driven decision-making through a rule-based recommendation engine while also exploring additional intelligent features for future development.

Service Recommendation Engine

The recommendation engine provides personalised salon service suggestions based on customer preferences.

When users create an account or update their profile, they can specify information such as:

- Hair type
- Face shape
- Hair length
- Preferred style
- Occasion

The backend analyses this information using predefined decision rules stored within the application.

For example:

Input

- Hair Type: Curly
- Face Shape: Oval
- Occasion: Formal

Recommended Services

- Elegant Curly Updo
- Defined Curl Styling
- Side-Swept Formal Curls

Another example:

Input

- Hair Type: Straight
- Face Shape: Round
- Occasion: Everyday

Recommended Services

- Long Layers
- Soft Waves
- Shoulder-Length Bob

The recommendation engine improves customer satisfaction by providing personalised suggestions that match individual preferences.

Chatbot (Future Enhancement)

To further improve customer support, the system proposes integrating a simple Natural Language Processing (NLP) chatbot.

The chatbot would answer frequently asked questions, including:

- Salon operating hours
- Service prices
- Stylist availability
- Appointment cancellation procedures
- Booking assistance

This feature would reduce staff workload while improving customer support.

Future Intelligent Features

Potential enhancements include:

- AI-powered face shape detection using uploaded photographs.
- Sentiment analysis of customer reviews.
- Predictive appointment scheduling.
- Machine learning algorithms that improve recommendations over time.
- Automated reminder notifications using customer behaviour analysis.

These enhancements demonstrate the system's potential for future expansion.

13. USER EXPERIENCE DESIGN AND RESPONSIVE INTERFACES

The TOUCHTASH application follows modern User Experience (UX) principles to ensure that the system is intuitive, attractive, and easy to use.

The interface has been developed using a mobile-first design approach with Bootstrap 5, allowing the application to adapt automatically to different screen sizes.

User Experience Principles

Simplicity

The interface is designed to minimise user effort by presenting only the information required for each task.

Consistency

Colours, fonts, icons, buttons, and navigation menus remain consistent throughout the application to improve usability.

Accessibility

The application includes:

- Clear navigation menus.
- Readable typography.
- Proper colour contrast.
- Responsive layouts.
- Keyboard accessibility where appropriate.

User Feedback

Users receive immediate feedback whenever they perform an action.

Examples include:

- Success notifications after booking appointments.
- Error messages for invalid input.
- Loading indicators while data is retrieved.
- Confirmation messages before deleting bookings.

Responsive Design

Bootstrap 5 ensures that every page automatically adjusts to different screen sizes including:

- Desktop computers

- Laptops
- Tablets
- Smartphones

This enables customers to access salon services from virtually any device.

Main User Interfaces

The application consists of several key pages:

Home Page

Introduces the salon and highlights featured services.

Services Page

Displays all available services with filtering and search functionality.

Booking Page

Allows customers to select services, choose stylists, and book appointments.

Customer Dashboard

Displays upcoming bookings, booking history, recommendations, and profile information.

Administrator Dashboard

Provides complete management functionality for customers, services, stylists, and appointments.

The overall interface has been designed to maximise usability while maintaining a modern and professional appearance.

14. SECURITY FEATURES

Security is a critical aspect of the TOUCHTASH application because it stores personal customer information and appointment records.

Several security measures have been implemented to protect both users and the application.

Password Encryption

User passwords are encrypted using bcrypt before being stored in the database.

This prevents passwords from being stored as plain text and significantly improves data security.

JWT Authentication

The application uses JSON Web Tokens (JWT) to authenticate users after login.

Authenticated users receive a secure token that is included in future requests when accessing protected resources.

This ensures that only authorised users can perform restricted operations.

Input Validation

All user input is validated before processing.

Validation checks include:

- Required fields
- Email format
- Password length
- Invalid characters
- Empty values

These checks reduce the risk of invalid or malicious input.

SQL Injection Prevention

Prepared SQL statements and parameterised queries are used to protect the database against SQL injection attacks.

Environment Variables

Sensitive information including:

- Database credentials
- JWT secret keys
- API configuration

is stored inside environment variables rather than directly in the source code.

Error Handling

The application provides user-friendly error messages while preventing sensitive system information from being displayed to users.

Detailed errors are logged internally for troubleshooting purposes.

These security practices improve the confidentiality, integrity, and reliability of the application.

15. DEPLOYMENT READINESS

The TOUCHTASH application has been developed with deployment in mind, making it suitable for hosting in a production environment.

Backend Deployment

The backend server can be deployed using platforms such as Render.

Deployment includes:

- Installing dependencies
- Configuring environment variables
- Starting the Express server
- Connecting to the production MySQL database

Frontend Deployment

The frontend can be deployed using Vercel or Netlify.

After deployment, users can access the application through any web browser.

Version Control

The project is maintained using GitHub.

GitHub provides:

- Version history
- Team collaboration
- Backup of source code
- Continuous development

Deployment Steps

To deploy the application:

1. Clone the GitHub repository.
2. Install project dependencies.
3. Configure environment variables.
4. Connect the application to the production database.
5. Start the backend server.
6. Deploy the frontend.
7. Verify that all APIs and database connections are functioning correctly.

Deployment Benefits

Deploying the application online allows:

- Remote customer access.
- Easy system maintenance.
- Real-world usability.
- Scalability for future growth.

The deployment-ready architecture ensures that TOUCHTASH can transition from an academic project into a practical business solution.

16. Entrepreneurship Opportunity

The TouchTash Intelligent Salon Booking System presents a strong entrepreneurship opportunity within the growing beauty and personal care industry. Many small and medium-sized salons still rely on manual booking methods such as phone calls, WhatsApp messages, or walk-ins, which can lead to double bookings, missed appointments, and poor customer experience.

TouchTash solves these problems by offering a centralized digital platform where customers can easily book appointments, while salon owners can manage schedules efficiently. The system can be commercialized as a Software-as-a-Service (SaaS) product, where salons subscribe monthly to use the platform.

Additional revenue opportunities include premium features such as automated reminders, customer analytics, marketing tools, and payment integration. This makes the system scalable and suitable for both small salons and larger beauty franchises.

17. Challenges and Reflection

During the development of the TouchTash system, several challenges were encountered. One of the main challenges was understanding how to integrate the front-end interface with the back-end database effectively, especially when handling booking data and user authentication.

Another challenge was ensuring that the system remains user-friendly while still supporting multiple features such as scheduling, notifications, and service selection. Balancing functionality and simplicity required careful design decisions.

From a learning perspective, this project improved our understanding of full-stack development, database management, and system design. It also strengthened problem-solving skills, especially when debugging errors and managing project deadlines.

18. Future Improvements

Although the current version of TouchTash meets the basic requirements, there are several improvements that can be made in the future to enhance functionality and user experience.

One major improvement would be the integration of online payment systems such as card payments or mobile money. Another enhancement would be the addition of a mobile application to make booking even more accessible.

The system can also be improved by adding artificial intelligence features such as personalized hairstyle recommendations based on customer history. Furthermore, implementing real-time notifications via SMS or email would improve communication between salons and customers.

Finally, expanding the platform to support multiple branches and advanced reporting dashboards would make it more suitable for large-scale businesses.

19. Conclusion

In conclusion, the TouchTash Intelligent Salon Booking System successfully demonstrates how technology can improve service delivery in the salon industry. It reduces manual workload, improves booking efficiency, and enhances customer satisfaction.

The project provided valuable experience in system development, database integration, and user interface design. Overall, TouchTash shows strong potential to be developed into a real-world commercial application that can benefit both salon owners and customers.



